

Venkata Reddy

Sacramento, CA | +1 (279) 667 5072 | venkatareddy01432@gmail.com

SUMMARY

Results-driven Data Engineer with 3+ years of experience in designing and optimizing scalable data pipelines for real-time and batch processing across diverse industries. Proficient in Big Data technologies like Apache Spark, Kafka, Airflow, and Databricks, along with cloud platforms such as AWS, GCP, and Azure for scalable data solutions. Skilled in SQL, Python, and Scala, with expertise in ETL automation, data modeling, and data governance, ensuring compliance with GDPR & HIPAA. Experienced in real-time data processing using Spark Streaming and Kafka, reducing latency and enabling actionable insights. Strong background in machine learning model integration using TensorFlow and Scikit-learn for predictive analytics. Adept at performance tuning and cost optimization, reducing infrastructure expenses while improving data pipeline efficiency.

SKILLS

Methodologies: SDLC, Agile, Waterfall	Cloud Technologies: AWS (S3, Redshift, EC2, Glue, EKS, Elastic Beanstalk), Azure (Data Lake, DataBricks), GCP
Programming Languages: Python, R, SQL, SAS	Databases: MySQL, PostgreSQL, SQL Server, Oracle
IDEs: Visual Studio Code, PyCharm, Jupyter Notebook	Version Control Tools: Git, GitHub, GitLab
Packages: NumPy, Pandas, Matplotlib, Seaborn, ggplot2, SciPy, Scikit Learn, TensorFlow	Other Skills: Data Cleaning, Data Wrangling, Critical Thinking, Communication Skills, Presentation Skills, Problem-Solving
Visualization Tools: Tableau, Power BI, Advance Excel, Grafana	Operating Systems: Windows, Linux, Mac
ETL Tools: SSIS, SSRS, Informatica, Talend	

EXPERIENCE

Oracle, USA Big data Engineer	April 2023 - Present
- Designed and implemented a scalable data pipeline for a SaaS platform, integrating Apache Spark, Kafka, and AWS S3 to process 2TB of daily data, enabling real-time analytics and improving data accessibility for cross-functional software teams. - Collaborated with software engineers to optimize real-time data processing using Spark Streaming, reducing latency by 30% and enabling faster feature rollouts for customer-facing applications. - Automated ETL workflows using Airflow and dbt, reducing manual intervention by 40% and improving data delivery speed for software development and testing teams. - Built a real-time fraud detection system using Kafka and Spark Streaming, reducing false positives by 15% and enhancing the accuracy of transactional data for a payment processing module. - Migrated on-premises data pipelines to AWS (S3, Glue, Redshift), reducing infrastructure costs by 25% and improving scalability for handling 1M+ events per second in microservices-based architecture. - Partnered with data scientists to deploy machine learning models into production, enabling real-time predictions and improving the personalization features of the software product. - Implemented data governance and access control policies to ensure compliance with GDPR and HIPAA, enhancing data security for customer data used in software development. - Ensured 99.9% data pipeline uptime by integrating Prometheus and Grafana for monitoring, minimizing disruptions, and enabling seamless data flow for continuous integration and deployment (CI/CD) pipelines.	
One Advanced, India Data Engineer	Aug 2019 - Nov 2021
- Engineered scalable ETL solutions using Apache Airflow, Python, and SQL, improving data workflow efficiency by 16% and enabling faster decision-making for business teams. - Led the migration of legacy systems from Oracle and SQL Server to Google BigQuery, achieving annual cost savings of \$678,000 and a 14% performance boost. Integrated Azure Data Lake and Databricks for advanced data processing and analytics. - real-time data pipelines using Apache Kafka and Spark Streaming to process and analyse streaming data, reducing latency by 40% and enabling real-time insights for business stakeholders. - Designed and maintained data pipelines to process 10+ TB of daily data using Apache Spark and Airflow, improving data ingestion speed by 67% and ensuring scalability for growing data volumes. - Developed interactive dashboards using Tableau and Power BI to visualize key metrics, enabling data-driven decision-making and improving operational efficiency by 29%. - Integrated machine learning models into data pipelines using TensorFlow and Scikit-learn, enabling predictive analytics and enhancing business forecasting accuracy by 22%. - Partnered with data scientists, software engineers, and business teams to design and implement 20+ data pipelines, ensuring alignment with business goals and technical requirements. - Optimized Spark jobs and SQL queries, achieving an average data loading rate of 1 GB/minute into BigQuery and improving overall system performance by 15%.	

EDUCATION

Master of Science in Computer Science University of Bridgeport, Bridgeport, CT	May 2023
Bachelor of Technology in Electronics and Computer Engineering KL University, Guntur, India	Apr 2020

CERTIFICATIONS

AWS Certified Developer Associate - Amazon
Snowpro Core Certification - Snowflake